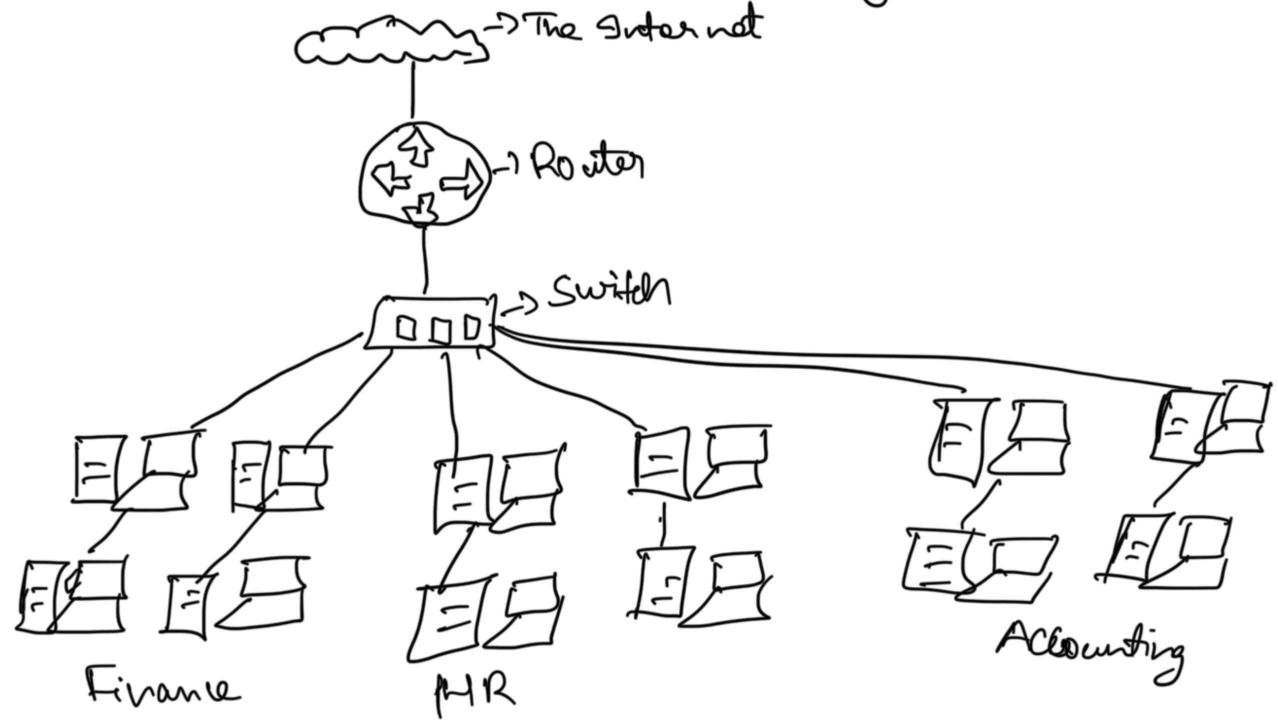
→ Task 2

A primer on Subnetting :-

As we discussed previously, A network can be found in All shapes and sizes—ranging from small to large. Subnetting is term given to splitting up network into smaller, maintain networks within itself. Ex:- Slicing up a cake for your friends. There is only a certain amount of cake to go around, but everybody wants a piece. Subnetting decides who get what slice & reserving such a slice of this metaphorical cake.

Take a business, for Example; you will different deparments such as:

1) Finance 2) Human Resource 3) Accounting



* You know where to send Information in real life to correct department. In the same way Network need to know as well. Network Administrators use subnetting to categorise and assign specific part of network to reflect this.

* Subnetting is achieved by splitting up number of hosts that

can fit within network, represented by a network called Subnet mask.

Octet#1 → 192 . 168 (octet 2) . 1 (Octet 3) . 1 → octet # 4

0-255 0-255 0-255 0-255

As we recap, IP address is made up of 4 sections called octet. The same goes for Subnet mask which is also represented as number of 4 bytes (32 bits). ranging from (0 - 255).

Subnets use IP Address in 3 different ways :-

1) Identify the network Address
2) Identify the host Address
3) Identify the default gateway.

1) Network Address :- purpose is this Address Identifies the start of actual network and is used to Identify a network's existence.

Example:- A device with IP Address of 192.168.1.100 will be on network identified by 192.168.1.0 → No single computer can use this address

2) Host Address :- An IP Address here is used to Identify a device on the Subnet.

Ex :- A device will have network address of 192.168.1.1 → This is unique ID of your device on specific network.

3) Default Gateway :- The default gateway address is special address assigned to device on network that is capable of sending information to another network.

Explanation :- Any data needs to go to device that isn't on the same network (i.e isn't on 192.168.1.0) will be sent on device. These devices can use host address but usually use either the first or last host address in network (.1 to .254)

Ex :- 192.168.1.254

Now in small networks such as at home, you will be on one subnet as there is unlikely chance that you need more than 254 devices connected at one time.
However this places such as business and offices will have much more of these devices (PC's, printers, cameras and sensors), where subnetting takes place.

Subnetting provides range of benefits, including :-
1) Efficiency 2) Security 3) Full control.
well come after how subnetting provides these benefits later date.

Ex :- lets take typical cafe on street. This cafe will have 2 networks.
1. One for employees, cash registers and other devices for the facility.
2. One for general public to use hotspot.

Subnetting allows you to seperate these two use cases from each other whilst having benefits of connection to larger networks such as Internet.

Quiz :-

1) what is the technical term for dividing a network up into smaller pieces?
A) Subnetting

2) How many bits are in subnet mask?
A) 32

3) what is the range of section (octet) of a subnet mask?
A) 0 - 255

4) what address is used to identify the start of a network?
A) Network Address

5) What address is used to Identify device within a network?

A) Host Address

6) what is name used to Identify the device responsible for sending data to another network?

A) Default Gateway.